Analog System Design Lab Mini-Project Report

Audio Preamplifier with Baxandall Bass and Treble Tone Control

Submitted by

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ABSTRACT

This project presents the design and simulation of an audio preamplifier incorporating a Baxandall bass and treble tone control network. The system is built around the TL072 dual low-noise operational amplifier, chosen for its superior audio performance characteristics over conventional op-amps. The Baxandall topology enables independent, symmetrical boost and cut of low-frequency (bass) and high-frequency (treble) components across the standard audio band of 20 Hz to 20 kHz, while maintaining a flat mid-frequency response.

The circuit operates from a dual supply of ±12 V and uses a passive RC Baxandall network followed by an active TL072-based amplifier stage. A potentiometer-based implementation allows continuous adjustment of bass and treble levels through parametric simulation in LTspice. The output stage is driven by an LM386 power amplifier IC, which provides sufficient current drive for small loudspeaker loads, making the system suitable for compact audio applications.

AC analysis simulations confirm the expected frequency response behaviour, demonstrating up to ±20 dB of independent control over bass and treble frequencies. The results validate the Baxandall topology as an effective and elegant tone control solution. The project illustrates key op-amp circuit design principles including frequency-selective feedback, active filtering, and cascaded gain stages in practical audio system design.

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1. INTRODUCTION

1.1 Problem Definition

This project addresses the design and simulation of an audio preamplifier with bass and treble tone control using the TL072 dual operational amplifier. The circuit is designed using the Baxandall tone control network, which allows the user to adjust the low frequency (bass) and high-frequency (treble) components of an audio signal independently and symmetrically. The circuit operates using a dual power supply of ±12 V and works across the full audio frequency range of 20 Hz to 20 kHz. A low-level AC audio signal is applied as the input, and the circuit provides an amplified output with adjustable bass and treble control. An LM386 power amplifier IC is included at the output stage to drive small loudspeaker loads. The expected outcome is that the output waveform should be amplified and the frequency response graph should show that the bass control boosts or cuts low frequencies while the treble control boosts or cuts high frequencies, with smooth variation depending on potentiometer settings.

1.2 Introduction

Music systems, televisions, mobile speakers, and other sound equipment make extensive use of audio amplifiers. In most audio systems, amplification of the signal alone is not sufficient. The quality of sound must also be adjusted to the listener's preference by increasing or decreasing the low-frequency (bass) or high-frequency (treble) components. A tone control circuit is used for this purpose.

In this project, an audio preamplifier with bass and treble tone control is designed and simulated using the TL072 low-noise dual operational amplifier. The TL072 is selected for its low noise characteristics and wide bandwidth, making it well suited for audio applications compared to general-purpose op-amps. The circuit is based on the Baxandall tone control network, one of the most widely used topologies in audio amplifiers due to its linear and symmetrical control of both bass and treble frequencies.

P. J. Baxandall introduced this tone control technique in 1952, and it became widely adopted owing to its ability to achieve effective frequency control using a simple resistor capacitor feedback network combined with an amplifier. The circuit operates by frequency-selective feedback: the amplifier gain is altered only in the low- or high-frequency range when the respective potentiometer is adjusted, while mid-frequencies remain virtually unaffected. The output of the tone control stage is fed into an LM386 power amplifier to provide sufficient drive for loudspeaker loads.

LTspice simulation software is used in this project to study the frequency response of the circuit under various bass and treble settings by plotting gain versus frequency, and to observe transient output waveforms at different audio frequencies.

1.3 Motivation

Tone control circuits are fundamental building blocks in virtually all consumer audio equipment. Understanding the design of a Baxandall tone control circuit provides insight into active filter design, frequency-selective feedback networks, and practical op-amp circuit design. The use of the TL072, a low-noise FET-input op-amp, reflects real-world audio design choices, and the addition of the LM386 power amplifier stage bridges the gap between a preamplifier and a complete audio system capable of driving a speaker

2. BACKGROUND THEORY AND OBJECTIVES

2.1 Background Theory

The Baxandall tone control circuit [1] is a passive RC network used in conjunction with an inverting amplifier. It achieves independent control of bass and treble frequencies through frequency-selective feedback. At mid-frequencies, the network presents equal impedance in both the feedback and input paths, resulting in unity gain. At low frequencies, the bass potentiometer shifts the ratio of feedback to input impedance, producing boost or cut. Similarly, at high frequencies, the treble potentiometer controls the gain in the upper audio band.

The TL072 is a dual JFET-input operational amplifier offering low noise, low harmonic distortion, and a wide gain-bandwidth product [2]. These characteristics make it superior to the LM741 or similar general-purpose op-amps for audio applications. Its slew rate and input impedance characteristics ensure that the tone control circuit performs faithfully across the full audio bandwidth without introducing significant distortion.

The LM386 is a low-voltage audio power amplifier designed for use in portable consumer electronic equipment such as radios and intercoms [3]. It provides a voltage gain of 20 to 200 and can deliver up to 250 mW into a 4 Ω load, making it suitable as an output driver stage following the Baxandall preamplifier. The combination of the TL072-based tone control and the LM386 power stage creates a complete audio amplification chain from signal source to loudspeaker.

Välimäki et al. [4] provide a comprehensive treatment of audio equalization techniques, establishing the theoretical foundations for filter-based tone controls. Their work confirms that the Baxandall topology remains one of the most practical implementations for parametric equalization in analog audio systems.

2.2 Objectives

- To design and simulate an audio preamplifier with Baxandall bass and treble tone control using the TL072 operational amplifier in LTspice.
- To verify that the circuit provides independent and symmetrical boost and cut of bass (low frequency) and treble (high frequency) components while maintaining a flat mid-frequency response.
- To integrate an LM386 power amplifier at the output stage and validate the complete signal chain from audio input to speaker-level output.

3. DESIGN AND METHODOLOGY

3.1 Block Diagram

The system consists of the following functional stages in

cascade: • Audio Input Source (mobile/AUX)

- Coupling Capacitor Stage (DC blocking)
- Buffer / Preamplifier Stage (TL072 – first op-amp section)
- Baxandall Tone Control Network (Active Filter with bass and treble potentiometers)
- Output Amplifier Stage (TL072 – second op-amp section)
- LM386 Power Amplifier Stage
- Audio Output Signal (amplified + tone adjusted) to loudspeaker

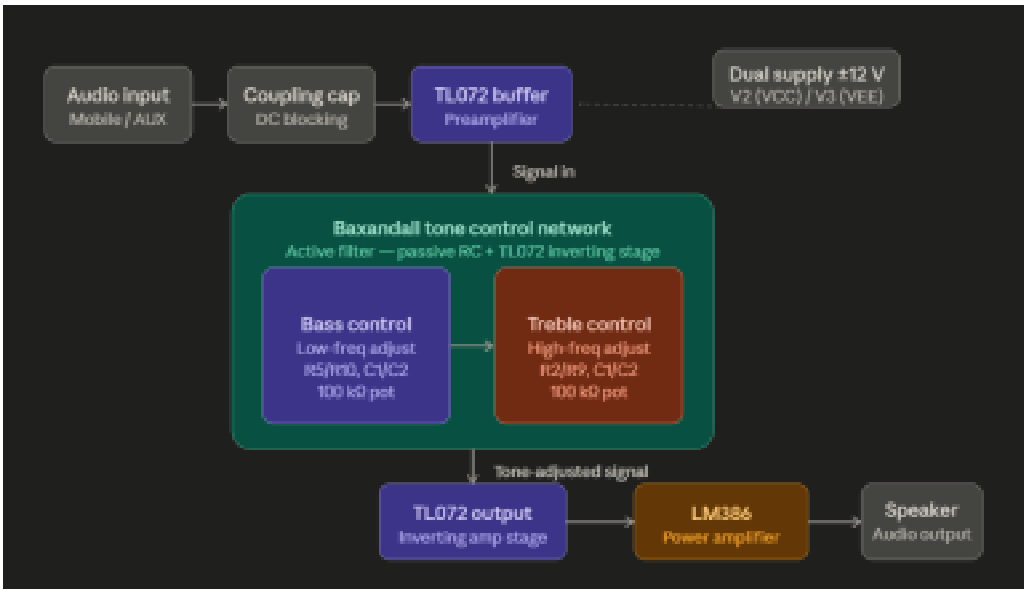

Figure 1. Block Diagram of the Audio Preamplifier with Baxandall Tone Control

3.2 Specifications

Table I: Circuit Specifications

Parameter	**Value / Description**
Power Supply	±12 V (dual supply)
Op-Amp (tone control & preamp)	TL072 (dual JFET-input)
Power Amplifier IC	LM386
Input Signal	Low-level AC audio (1 V AC, 1 kHz reference)
Frequency Range	10 Hz – 100 kHz
Bass Potentiometer (R5/R10)	100 kΩ (parameterised: bass = 0–1)
Treble Potentiometer (R2/R9)	100 kΩ (parameterised: treble = 0–1)
Load Resistor (R8)	10 kΩ (simulation load)
Simulation Tool	LTspice – AC analysis (.ac dec 100 10 100k)

3.3 Design and Implementation

3.3.1 Baxandall Tone Control Network

The Baxandall tone control network consists of two passive RC sections – one controlling bass frequencies and the other controlling treble frequencies – connected to an inverting amplifier built around the TL072. The bass section uses capacitors C1 and C2 (22 nF each) in combination with the bass potentiometer (R5/R10, 100 kΩ total). At low frequencies, these capacitors present high impedance, and the potentiometer wiper position determines the ratio of feedback to input signal, thus controlling bass gain. The treble section uses capacitors in parallel with the treble potentiometer (R2/R9), which become dominant at high frequencies, similarly controlling the treble gain.

Component R9 (27 kΩ) sets the mid-frequency gain reference, while R1 and R4 (5.6 kΩ each) provide input and feedback termination resistors. Capacitor C3 (1.8 nF) across R9 provides frequency compensation to prevent high-frequency instability. The input resistors R5 and R8 (3.3 kΩ each) define the input impedance of the bass section. This passive network is driven from the non-inverting buffer input of the TL072 and the output is taken from the inverting amplifier configuration.

3.3.2 TL072 Op-Amp Stage

The TL072 dual op-amp is configured as an inverting amplifier providing the active gain for the tone control network. The dual supply of ±12 V ensures sufficient headroom for the audio signal. The TL072 offers a typical gain-bandwidth product of 3 MHz and a slew rate of 13 V/μs, making it suitable for the full audio bandwidth. Its low input noise voltage (~18 nV/√Hz) ensures that the preamplifier stage does not significantly degrade the signal-to-noise ratio.

3.3.3 LM386 Power Amplifier Output Stage

The LM386 power amplifier IC is connected at the output of the TL072 tone control stage. It provides a voltage gain of 20 (26 dB) in its default configuration and can deliver up to 250 mW into a 4 Ω to 8 Ω speaker load from a single supply voltage of 5–12 V (or from the positive rail of the dual supply). A bypass capacitor on pin 7 reduces power-supply ripple, and an output coupling capacitor blocks DC from the speaker. The LM386 thus serves as the power output driver, completing the audio amplification chain.

3.3.4 Circuit Schematic

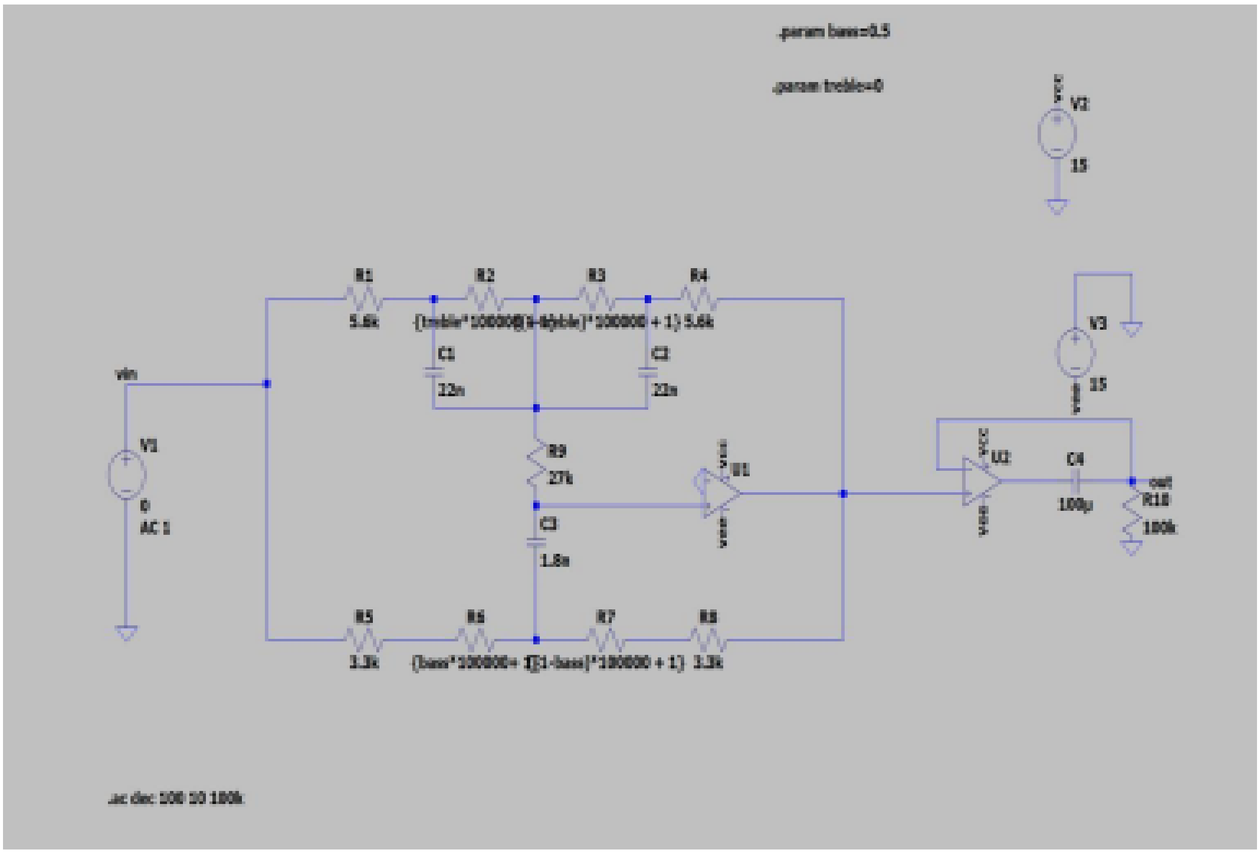

Figure 2. LTspice Schematic of the Baxandall Tone Control Circuit with TL072 and LM386

The circuit shown in Figure 2 implements the complete Baxandall tone control schematic simulated in LTspice. The parametric values {treble*100k+1m} and {bass*100k+1m} allow the potentiometer positions to be swept during simulation by varying the .param treble and .param bass values between 0 and 1. The AC analysis command .ac dec 100 10 100k sweeps from 10 Hz to 100 kHz with 100 points per decade, providing high resolution frequency response data.

4. RESULT ANALYSIS

4.1 Simulation and Testing

The circuit was simulated in LTspice using the AC analysis command to obtain the frequency response under various tone control settings. The .param directives allow the bass and treble potentiometer positions to be set from 0 (fully cut) to 1 (fully boost), with 0.5 representing the flat (unity gain) position at mid-frequencies.

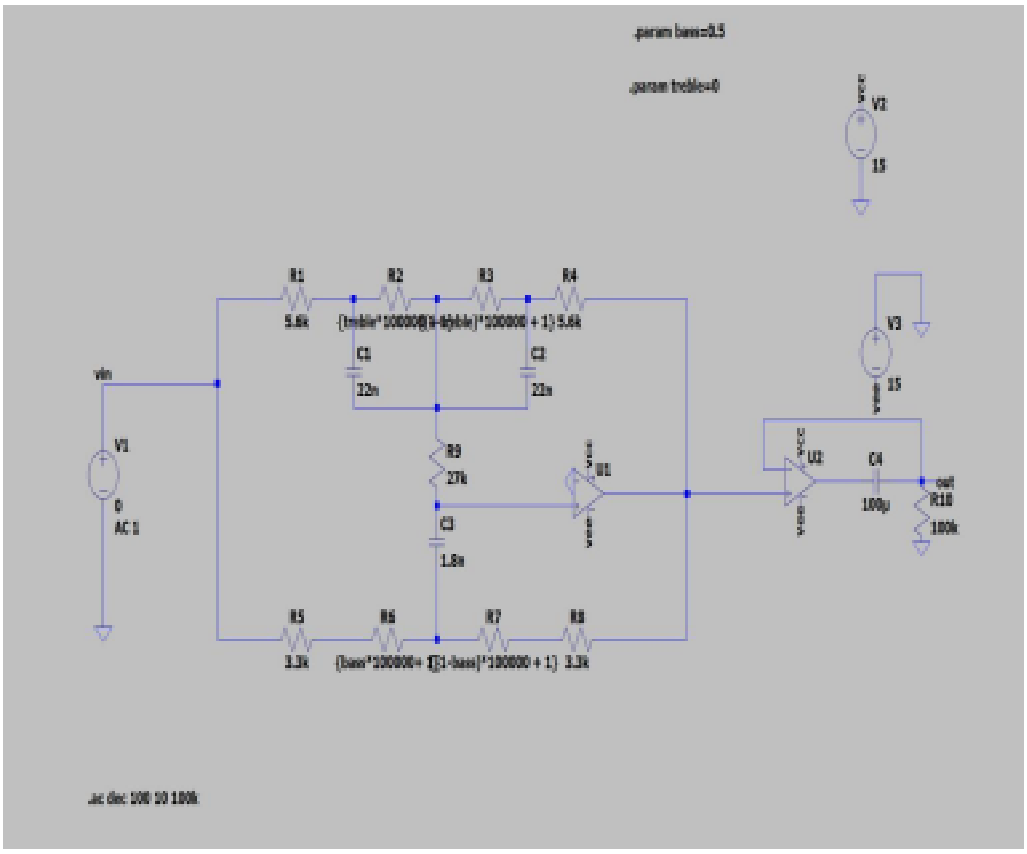

Figure 3. LTspice Circuit for AC Analysis

4.1.1 Simulation Result for Bass Control

(i) Bass cut, treble centred (.param treble = 0.5; .param bass = 0)

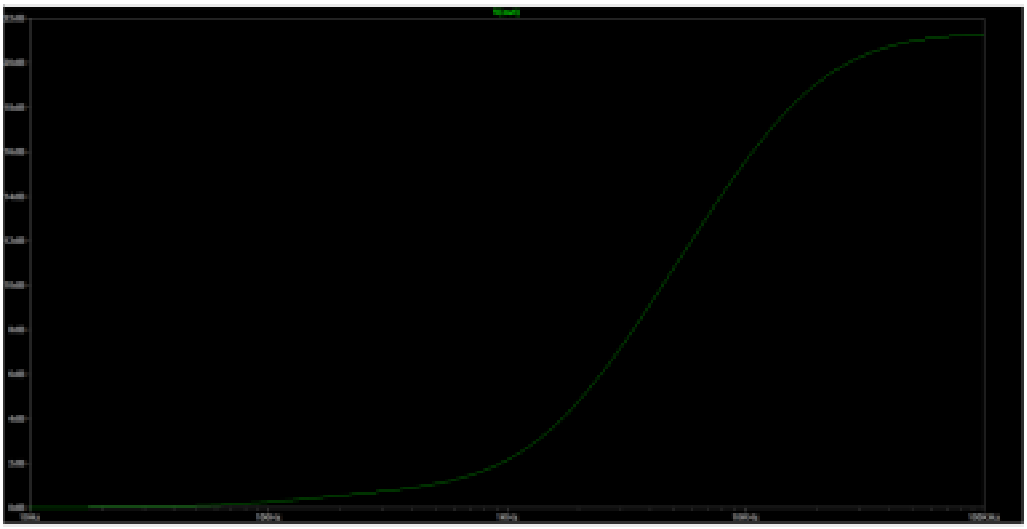

Figure 4. Frequency Response — Bass Cut (.param bass = 0, treble = 0.5)

The frequency response shows attenuation at low frequencies, confirming effective bass cut due to the RC network reducing gain below the corner frequency. At higher frequencies, the response remains nearly unchanged, indicating the treble control is set at its midpoint and does not introduce boost or cut. This behaviour validates proper operation of the Baxandall tone control, where bass and treble sections act independently to shape the overall response.

(ii) Bass boost, treble centred (.param treble = 0.5; .param bass = 1)

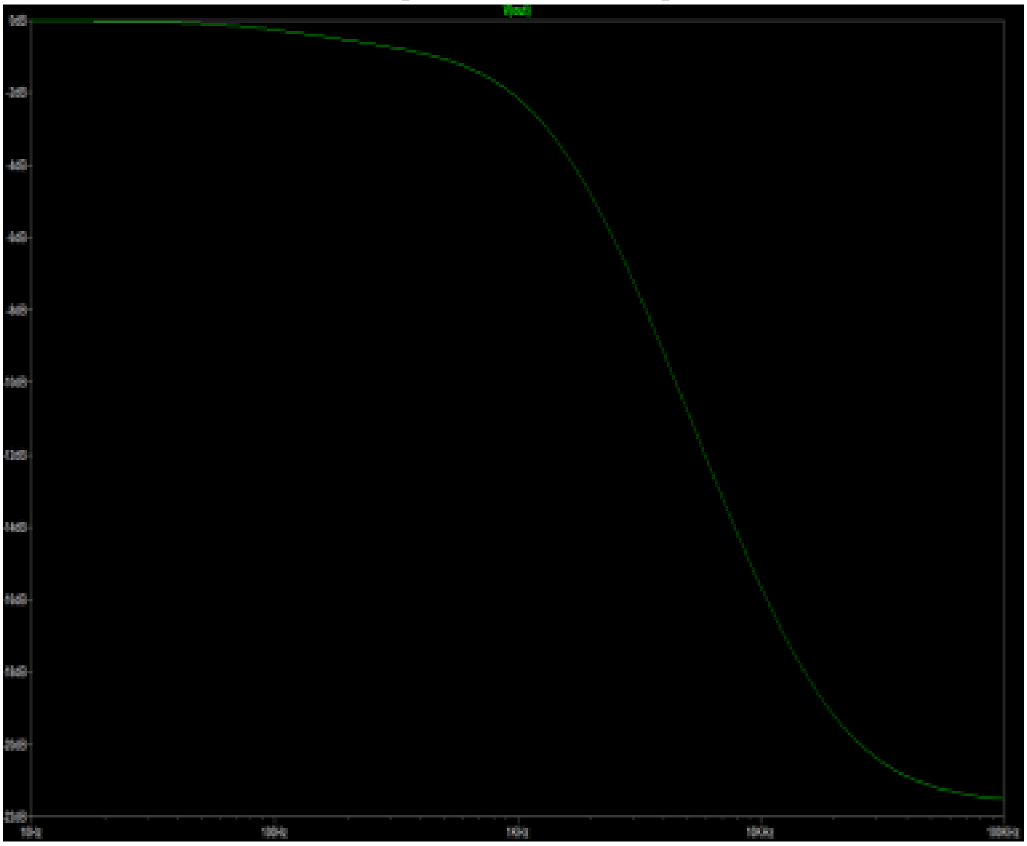

Figure 5. Frequency Response — Bass Boost (.param bass = 1, treble = 0.5)

The response exhibits increased gain at low frequencies, confirming effective bass boost due to the RC network enhancing signals below the cutoff frequency. At higher frequencies, the gain rolls off and remains unaffected, indicating the treble control is maintained at its midpoint without introducing additional boost or attenuation.

4.1.2 Simulation Results for Treble Control

(i) Treble cut, bass centred (.param treble = 0; .param bass = 0.5)

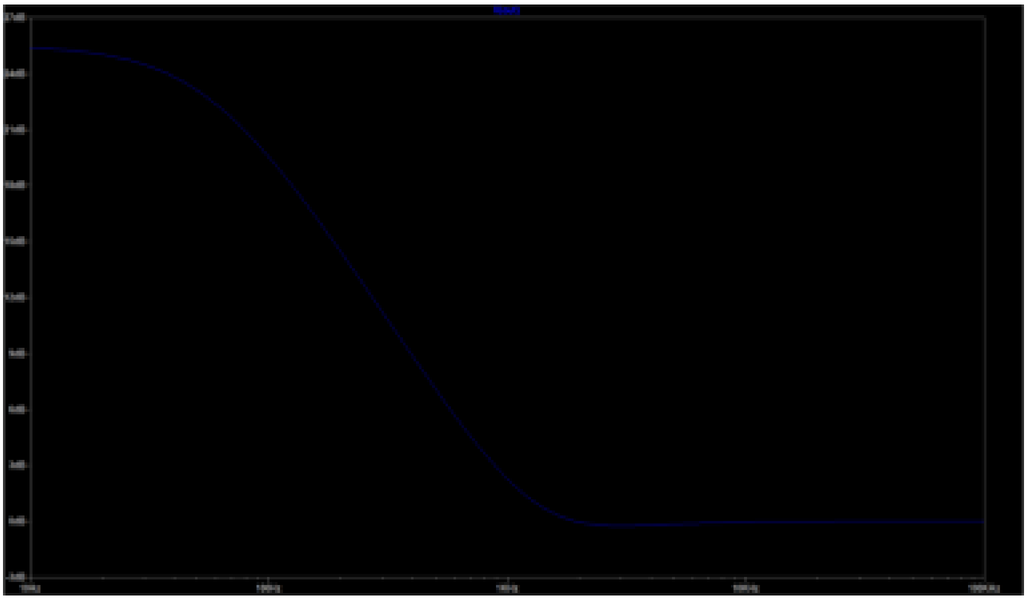

Figure 6. Frequency Response — Treble Cut (.param treble = 0, bass = 0.5)

The frequency response shows significant attenuation at higher frequencies, confirming effective treble cut due to the RC network reducing gain above the cutoff frequency. At lower frequencies, the gain remains nearly constant, indicating the bass control is set at its midpoint and does not introduce boost or attenuation.

(ii) Treble boost, bass centred (.param treble = 1; .param bass = 0.5)

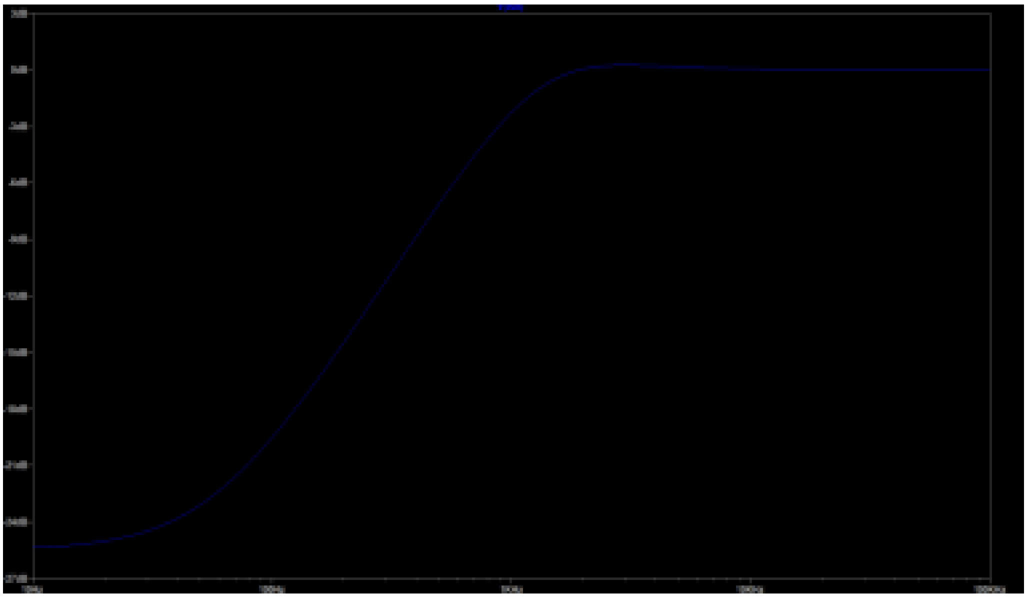

Figure 7. Frequency Response — Treble Boost (.param treble = 1, bass = 0.5)

The frequency response shows increased gain at higher frequencies, confirming effective treble boost due to the RC network enhancing signals above the cutoff frequency. At lower frequencies, the gain remains nearly constant, indicating the bass control is set at its midpoint and does not introduce boost or attenuation.

4.1.3 Treble Control — Combined Response

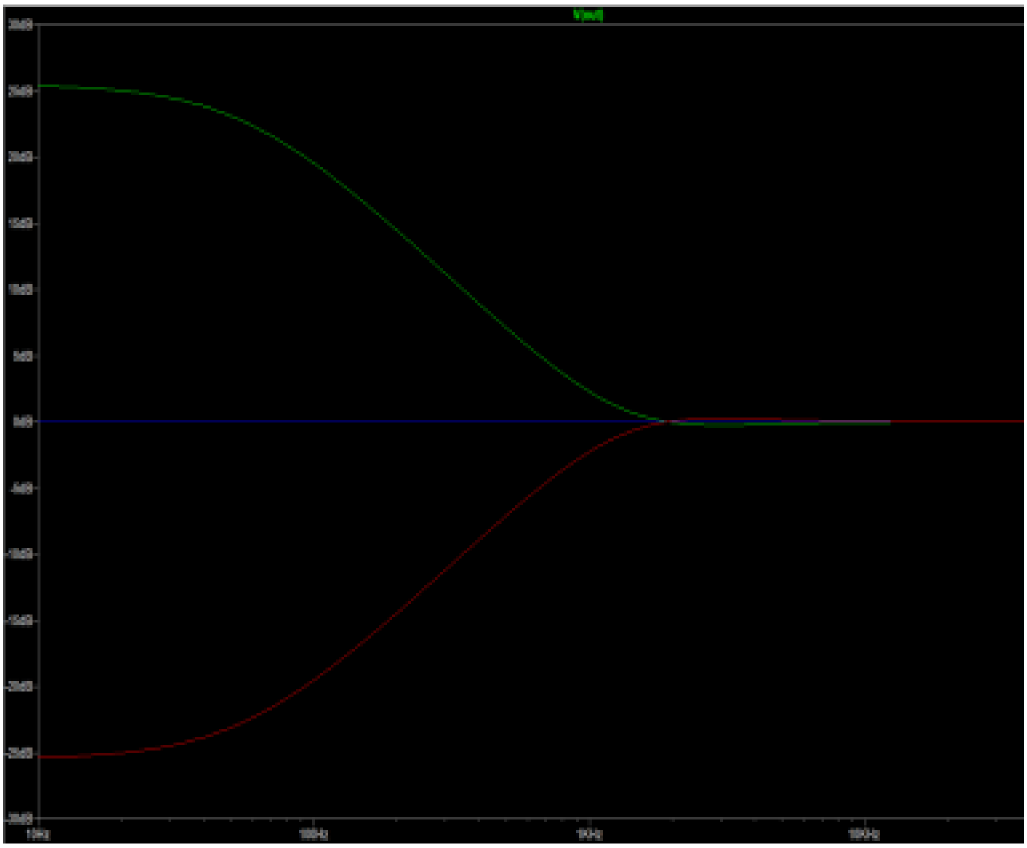

Figure 8. Treble Control — Combined Boost and Cut Response

4.1.4 Bass Control — Combined Response

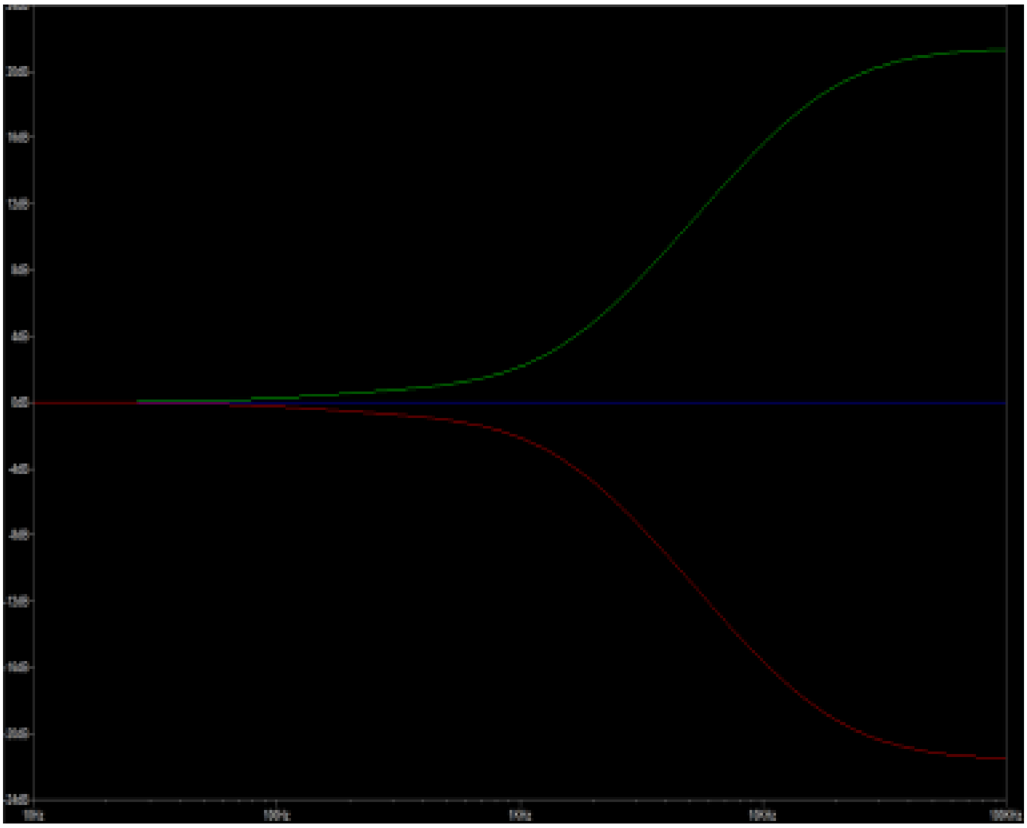

Figure 9. Bass Control — Combined Boost and Cut Response

5. CONCLUSION AND FUTURE SCOPE

5.1 Summary

This project successfully demonstrates the design and LTspice simulation of an audio preamplifier with Baxandall bass and treble tone control. The TL072 dual op-amp provides low-noise active amplification, and the passive Baxandall RC network enables independent symmetrical control of bass and treble frequencies. The LM386 power amplifier output stage provides practical speaker-level drive capability.

5.2 Conclusion

The simulation results confirm that the Baxandall tone control circuit achieves the design objectives: independent bass and treble boost/cut. The TL072 is validated as an appropriate choice for audio preamplifier design due to its low noise and wide

bandwidth. The cascaded LM386 power stage extends the circuit into a complete audio amplification system. The project reinforces fundamental analog design concepts including frequency
selective feedback, active filter topologies, and cascaded op-amp gain stages.

5.3 Future Scope

The circuit can be extended to include a mid-frequency (presence) control, creating a three-band equaliser. A hardware prototype on a printed circuit board using TL072 and LM386 ICs would validate the simulation results in practice. Additionally, the potentiometers could be replaced with digitally controlled resistors (DCRs) driven by a microcontroller to implement a digitally programmable tone control system. Further study could explore the effect of op-amp bandwidth limitations on the high-frequency treble response and methods to compensate for them.

REFERENCES

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ANNEXURES

Appendix A: Component Datasheets

Datasheets for the TL072 dual op-amp, LM386 audio power amplifier, and passive components used in the circuit are to be appended here.

https://www.ti.com/lit/ds/symlink/tl072.pdf

https://www.ti.com/lit/ds/symlink/lm386.pdf